

Jorge Hernández Galeano

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PROFILE

PhD candidate in medical imaging (Université Paris-Saclay / GE HealthCare) working at the intersection of **X-ray physics, inverse problems and deep learning**. Two years of industrial research on spectral breast imaging, from physics-based forward modeling to learned lesion characterization. Strong background in signal processing and estimation theory and hands-on experience with model-based deep learning for tomographic reconstruction.

EDUCATION

- **Ph.D. in Medical Imaging** Villejuif, France
Université Paris-Saclay, BioMaps Laboratory 2025 – 2028 (expected)
 - **Thesis:** *Spectral X-ray Breast Imaging for Lesion Characterization*
 - Industrial PhD (CIFRE) in collaboration with GE HealthCare
 - Topics: physical modeling, material decomposition, signal and image processing, machine learning
- **Engineering Diploma (Master's Degree) – Signal Processing and Artificial Intelligence** Cergy-Pontoise, France
École Nationale Supérieure de l'Électronique et de ses Applications (ENSEA) 2022 – 2024
 - **Ranked 5/28 | Master's thesis graded 18/20 with highest honors** (GE HealthCare): *Deep Learning-Based Differentiation of Lesions in Mammography Using Compositional Imaging*
 - **Coursework:** estimation and detection theory, stochastic signal processing, advanced signal processing, information theory and compression, machine learning, deep learning
- **Electronic Engineering** Bogotá, Colombia
Universidad Nacional de Colombia 2019 – 2024
 - **Highest honors** (2019–2022); competitively selected for transfer to the ENSEA engineering program (Grande École)
 - **Coursework:** electromagnetic fields, circuit analysis, signal processing, communication systems, electronics, electromagnetic compatibility

EXPERIENCE

- **Doctoral Research Engineer (CIFRE)** Buc, France
GE HealthCare April 2025 – Present
 - **Lead the research project** on dual-energy X-ray material decomposition for breast lesion characterization, jointly with the BioMaps laboratory; own the full pipeline from physical modeling to evaluation
 - Bridge academic research and industrial product development, presenting results to GE HealthCare R&D teams
- **Research Intern – Spectral X-ray Imaging and Deep Learning** Buc, France
GE HealthCare January 2024 – July 2024
Academic referent: Guillaume Renton
 - Designed and built a **physics-based simulation framework** for dual-energy mammography (energy-dependent attenuation, tissue composition), producing **40,000+ synthetic acquisitions** and enabling studies that would be infeasible with clinical data alone
 - Implemented X-ray material decomposition and trained a CNN reaching **0.99 AUC** for lesion characterization on simulated data; validated robustness across tissue compositions and interpreted the model with Layer-CAM
 - Work led to the CIFRE PhD position with GE HealthCare
- **Research Intern – Computer Vision and Deep Learning** Cergy, France
CNRS ETIS Laboratory May 2023 – August 2023
Supervisors: Camille Simon Chane, Aymeric Histace
 - Tackled training-data scarcity in biological imaging with **GAN-based data augmentation**: reviewed the literature, then implemented and benchmarked architectural variants (residual blocks, spectral normalization, Wasserstein loss with gradient penalty)
 - Generated realistic synthetic images with **FID below 13**, validated with IS, SSIM and t-SNE analysis of feature distributions

RESEARCH PROJECTS

- **Deep Learning for Sparse-View Computed Tomography Reconstruction** [🔗]

October 2023 – December 2023

Supervisor: *Ishak Ayad*

- Implemented the **LEARN unrolled reconstruction network** from scratch in PyTorch, coupling a CT forward model (ODL/ASTRA) with learned CNN regularizers so that each layer performs one iteration of a gradient-based reconstruction algorithm
- Characterized reconstruction quality under increasingly sparse projection views, a core challenge of limited-data inverse problems; code publicly available

RESEARCH INTERESTS

Inverse problems in computational imaging | Physics-informed machine learning | Unrolled and model-based deep learning | Reconstruction under limited measurements | Forward modeling for simulation-based inference | Spectral X-ray methods

TECHNICAL SKILLS

- **Languages:** Python, MATLAB, C++
- **Machine learning:** PyTorch, PyTorch Lightning, scikit-learn, Weights & Biases, Optuna
- **Scientific computing:** NumPy, SciPy, Pandas, Polars, ODL, ASTRA Toolbox
- **Research:** Physics-based forward modeling, simulation study design, statistical validation, scientific writing

LANGUAGES

English (C1) | French (B2) | Spanish (Native)